



Latest Advocacy Update Webinar

Congressional Briefing on kratom

REQUEST TO FDA FOR STAKEHOLDERS MEETING

The FDA has waged a persistent war on kratom, including making unsupported claims in official publications in the Federal Register that kratom's main alkaloids pose an imminent hazard to public safety.

It is our sincere hope that Commissioner Califf will see the importance of a public stakeholder meeting where conflicts in the Agency's policy on kratom and the current science and policy landscape can be fully evaluated.

[Read full letter here](#)

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From: McCurdy,Christopher R
Sent time: 10/02/2024 02:16:26 PM
To: staylor@americankratom.org
Subject: Re: slides from briefing
Attachments: Kratom US Congressional Update McCurdy Sept 2024.pdf Kratom US Congressional Update McCurdy Sept 2024.pdf

Please find the slides attached here.

Christopher R. McCurdy, Ph.D., F.A.A.P.S.
Associate Dean for Faculty Development
Professor of Medicinal Chemistry
The Frank A. Duckworth Eminent Scholar Chair
Director, UF Translational Drug Development Core
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From: staylor@americankratom.org <staylor@americankratom.org>
Sent: Thursday, September 26, 2024 8:51 AM
To: McCurdy,Christopher R <CMccurdy@cop.ufl.edu>
Subject: slides from briefing

[External Email]

Good Afternoon Dr. McCurdy,

When you have some time, can you please send me your Congressional Briefing slides and written statement?

Thank you for your help,



Stephanie Taylor
Research and Project Assistant, The American Kratom Association

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Science Update on *Mitragyna speciosa* **KRATOM**

Christopher R. McCurdy, BScPh, PhD, FAAPS
The Frank A. Duckworth Eminent Scholar Chair
Associate Dean for Faculty Development
Professor of Medicinal Chemistry & Pharmaceuticals
College of Pharmacy, University of Florida
Director, UF CTSI Translational Drug Development Core
AAPS President, 2017-2018



Kratom = Threat or Therapeutic

- Anecdotally used for chronic pain, mood elevation, opioid use disorder, alcohol use disorder
- Adulterated or contaminated products reported
- Claims that kratom is severely addictive and deadly
- World Health Organization / US Food and Drug Administration
 - Ongoing talks regarding therapeutic vs. harm profiles



Kratom illegal in 4 states (Alabama, Arkansas, Indiana, Wisconsin)

Kratom Consumer Protection Act made law in 8 states with several pending

Acknowledgements

- **Chemistry**

- Francisco Leon, PhD
- Marco Mottinelli, PhD
- Sushobhan Mukhopadhyay, PhD
- Sampa Gupta, PhD
- Grant Zwolinski, PharmD
- Ahmed Awad, PhD
- Nelson Chear (USM)
- Nicholas Guadagnoli

- **Receptor Binding and Function**

- Samuel Obeng, PhD
- Takato Hiranita, PhD
- Maria Seabra, PhD
- Clinton Canal, PhD (Mercer University – 5HT)
- Yiming Chen, PhD (Mercer University)

- **DMPK**

- Bonnie A. Avery, PhD
- Abhisheak Sharma, PhD
- Shyam Kamble, PhD
- Raju Kanumuri, PhD
- Tamara King, PhD
- Erin Berthold, PhD
- Michelle Kuntz
- Alexandria Senetra, MS

- **Behavioral Pharmacology**

- Lance McMahon, PhD (now at TTUHSC)
- Takato Hiranita, PhD
- Jay McLaughlin, PhD
- Jenny Wilkerson, PhD (now at TTUHSC)
- Lisa Wilson, PhD
- Yuma Ortiz
- Scott Hemby, PhD (Highpoint Univ)
- Jonathan Katz, PhD (NIDA IRP)

- **Horticultural Science**

- Brian Pearson, PhD (UF/IFAS)
- Satya Swathi Nadakuduti, PhD (UF/IFAS)
- Sarah O'Conner (Max Planck, Germany)

- NIDA UG3/UH3 DA048353

- NIDA R01 DA047855

- Urban Ice Organics and Kelly Dunn

- Scott Akers and Jeremy Steding

- University of Florida Foundation

- Multiple UF student workers!

Mitragyna speciosa

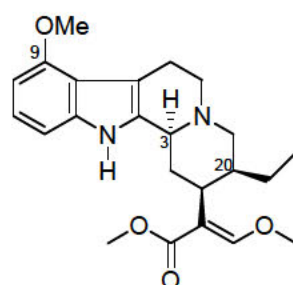
- FAMILY: Rubiaceae
- GENUS: *Mitragyna*
- SPECIES: *speciosa*
- Tree found in tropical Southeast Asia, particularly Thailand and Malaysia.
- Referred to as “Kratom” in Thailand and “Biak Biak or Ketum” in Malaysia.
- Contains over 40 alkaloids that have been isolated to date.¹



¹ Adkins, J.E.; Boyer, E.W.; McCurdy, C.R. *Curr. Topics Med. Chem.*, **11**, 1165-75 (2011)

Kratom Use in Southeast Asia

- Kratom tea is used by field workers in **Malaysia** to relieve **pain**, as a **stimulant** to improve work capacity, and **to reduce opioid withdrawal**¹
- Leaves are **CHEWED** in **Thailand**
- Most kratom raw material **in USA** is from **Indonesia**
- The predominant active agent in kratom is **mitragynine (MG)**



Mitragynine



¹ Jansen K.L.R., Prast C.J. *J. Ethnopharmacology*. **23**, 115-119 (1988)

² Singh, D. et al. *J. Ethnopharmacology*. **249**, 112462 (2020)

Kratom Use in USA

- Widely available across the internet and smoke/vape shops
- Not federally regulated or really monitored
- Current estimates in the USA **>15 million users**



ADDICTION

SSA | SOCIETY FOR THE
STUDY OF
ADDICTION

LETTER TO THE EDITOR |  **Free Access**

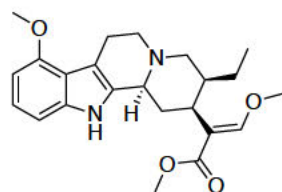
Not all kratom is equal: The important distinction between native leaf and extract products

Oliver Grundmann  Albert Garcia-Romeu,
Christopher R. McCurdy, Abhisheak Sharma, Kirsten E. Smith,
Marc T. Swogger, Stephanie T. Weiss

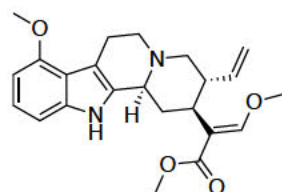
First published: 09 October 2023 |
<https://doi.org/10.1111/add.16366>

UF | College of Pharmacy
UNIVERSITY of FLORIDA

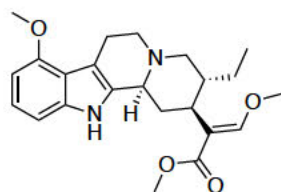
Isolation of kratom alkaloids



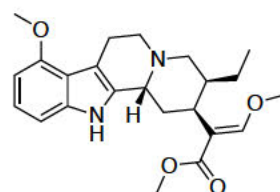
Mitragynine
66%
0.7-38.7%



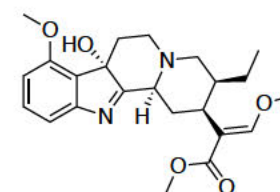
Paynantheine
9%



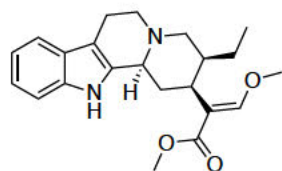
Speciogynine
7%



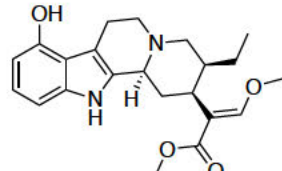
Speciociliatine
~1%
~10%



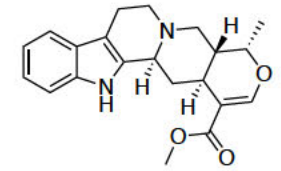
7α-hydroxymitragynine
~2%
<0.01%



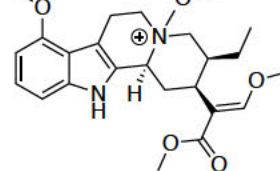
Corynantheidine
<1%



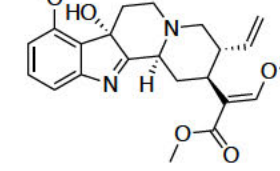
9-Hydroxycorynantheidine
<1%



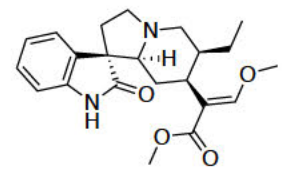
Ajmalicine
<1%



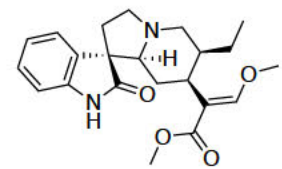
Mitragynine N-oxide
<1%



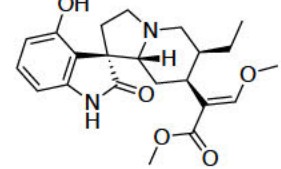
7α-hydroxypaynantheine



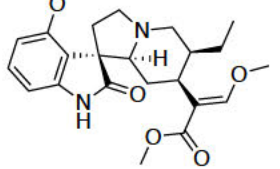
Corynoxine A
<1%



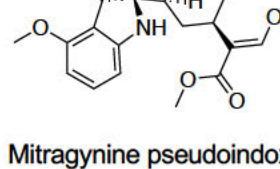
Corynoxine B
<1%



Isospeciofoline
<1%



Mitragynine oxindole B
<1%

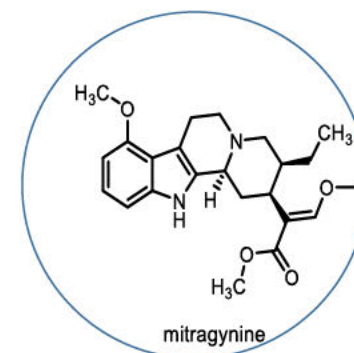


Mitragynine pseudoindoxyl

Binding and functional effects of mitragynine

- Mitragynine has **partial agonist effects** at the μ opioid, α_{1A} and α_{1D} adrenergic receptors
- Poison Control Centers: Kratom overdoses resemble **stimulants** (not opioids)

Target	Ki (nM)	Agonist/ Antagonist
μ	136.0	Partial agonist
κ	157	Antagonist
δ		Antagonist
α_{1A}	1,660	Partial agonist
α_{1B}	2,490	Antagonist
α_{1D}	4,610	Partial agonist
α_{2A}	3,590	ND
α_{2B}	9,190	ND
α_{2C}	1,400	ND



Obeng, Samuel, et al. *J Med Chem.* 63 (2019). 433-439.

Therapeutic Potential of Kratom

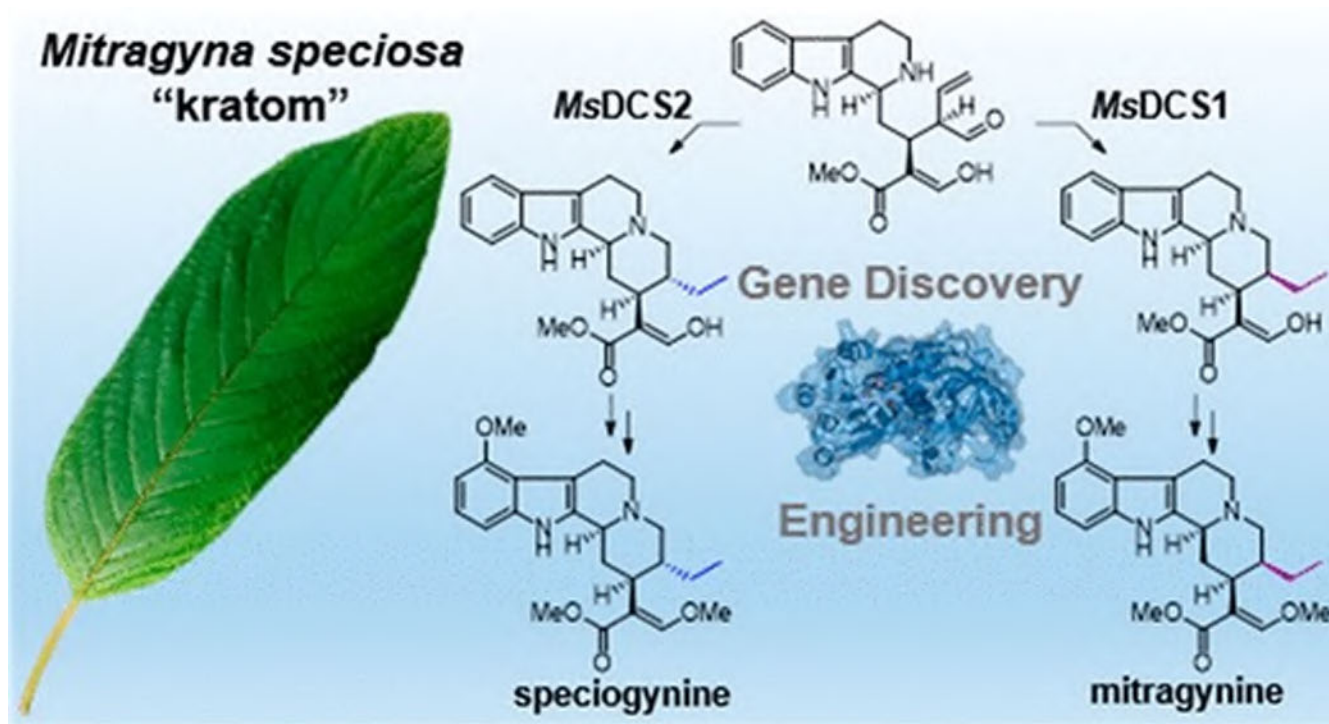
- **Opioid Detoxification:** Kratom has potential to replace several medications used during detoxification (**opioid, adrenergic, analgesic and anxiolytic**). This would improve medication adherence and chances of completing detoxification.
- **Medication Assisted Therapy:** Kratom is informally used to reduce opioid use. Kratom withdrawal is mild (<9 on SOWS scale). **Polydrug users report Kratom also reduces methamphetamine use.**
- **THE LACK OF A STANDARDIZED PRODUCT HAS DELAYED RIGOROUS CLINICAL TRIALS TO EVALUATE THESE CLAIMS**

Mitragyna speciosa growing at the University of Florida MREC



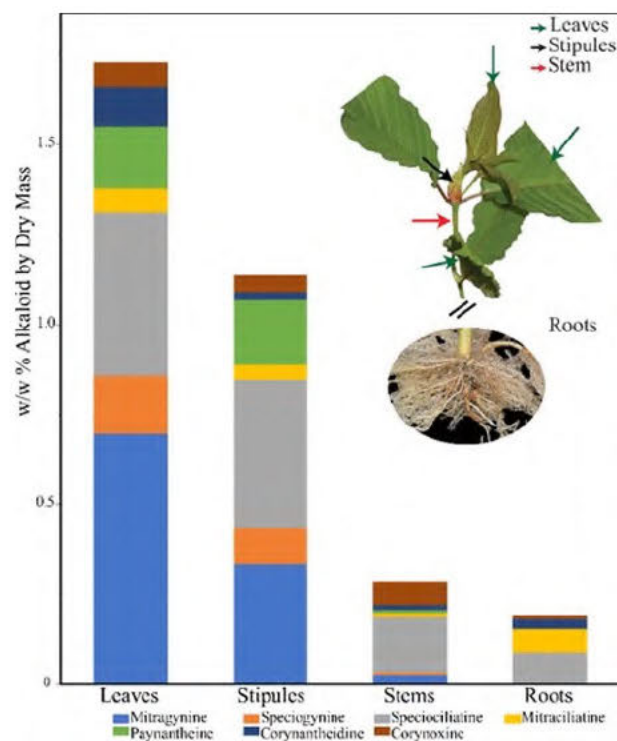
Professor Brian Pearson, UF IFAS

Mitragynine Biosynthetic pathway deciphered

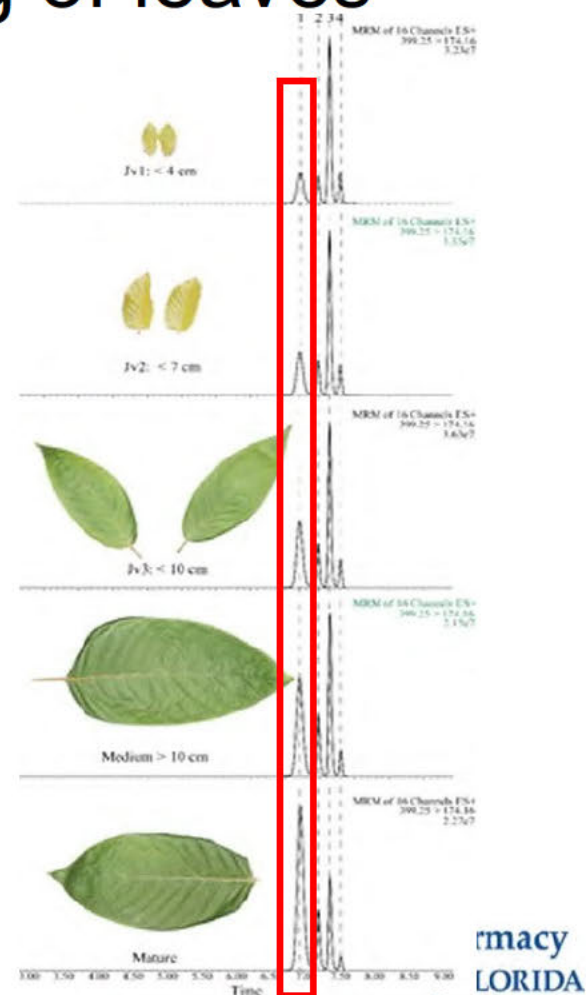


Schotte C., et al., *J. Am. Chem. Soc.* 2023 Mar 8, 145, 9, 4957–4963

Developmental alkaloid profiling of leaves

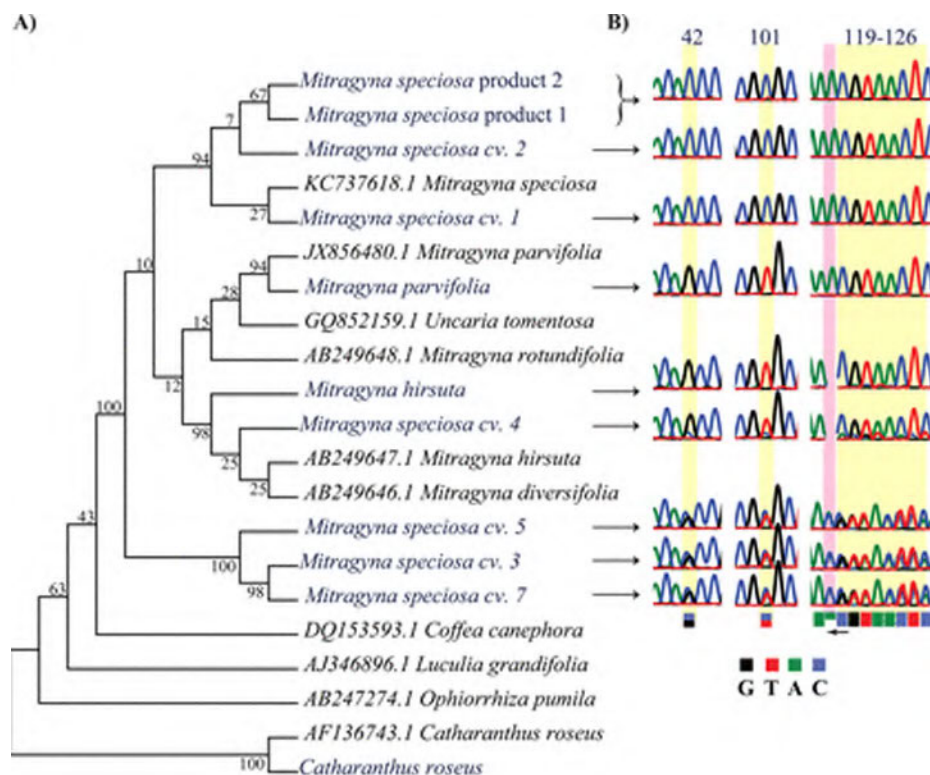


- Mitragynine is also present in stipules
- Mitragynine content increases with leaf development

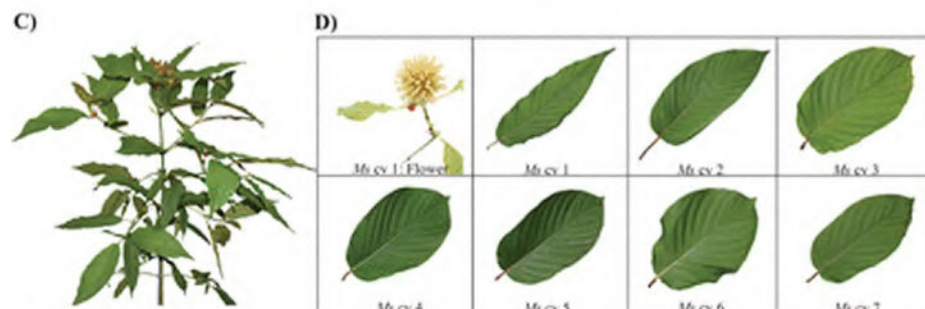


Laforest LC, et al., *J. Nat. Prod.* 2023 Mar 13

macy
LORIDA



DNA barcoding and phylogenetic analysis of *Mitragyna* cultivars



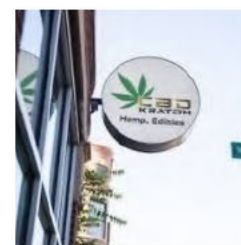
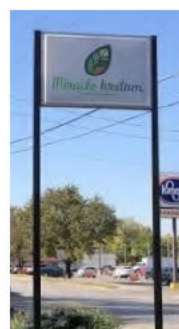
Laforest LC, et al., *J. Nat. Prod.* 2023 Mar 13



Alkaloidal content in 211 commercial kratom products available in the USA



Product Content % (w/w%)										
Alkaloid	Mitragynine	7-Hydroxymitragynine	Speciogynine	Speciociliatine	Mitraciliatine	Mitraphylline	Paynantheine	Corynantheidine	Corynoxine	Corynoxine B
Range	0.27-1.72	BLLOQ-0.03	0.04-0.34	0.16-0.79	0.01-0.11	BLLOQ	0.05-0.45	0.01-0.07	BLLOQ-0.18	BLLOQ-0.07
Mean $\pm$ S.D.	0.79 $\pm$ 0.21	-	0.16 $\pm$ 0.05	0.33 $\pm$ 0.09	0.04 $\pm$ 0.01	-	0.21 $\pm$ 0.06	0.02 $\pm$ 0.01	-	



UPLC-PDA-Q-TOF finger printing of authentic kratom samples (left) and 7- Ω HMZ tablet (right)
(Several unknown compounds were identified in 7- Ω HMZ tablet, and 7-hydroxymitragynine (7-HMG) was not present in kratom leaf sample.)

